Suman Kumar

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Master of Computer Applications
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GitHub Profile
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Education

- **National Institute of Technology, Warangal** *2024–2027*
  Master of Computer Applications (MCA)
- **Tilka Manjhi Bhagalpur University** *2018–2022*
  Bachelor of Computer Applications (BCA)
- **Bihar School Examination Board** *2016–2018*
  Intermediate
- **Bihar School Examination Board** *2015–2016*
  High School

Publications

- **Identity Elimination in EML Trees** *2026*
  Co-Author | Dept. of CSE, NIT Warangal (Supervisor: Dr. Sumit Mishra)
  - Designed a **bottom-up tree rewriting algorithm** over abstract syntax trees, using structural pattern matching to detect and collapse inverse-pair compositions (e.g., $\ln(e^x) \to x$, $e^{\ln x} \to x$) in $O(n)$ traversal of n-node trees.
  - Proved **termination and confluence** of the rewrite system via fixpoint iteration, guaranteeing a unique, maximally reduced normal form for any input tree regardless of rewrite order.
  - Formulated **K-Count**, a formal tree-complexity metric combining symbol count and structural depth, achieving a **22.0% reduction** in AST nodes and **22.2% reduction** in operator calls across 33 benchmark functions.
  - Established **functional completeness** of a single binary operator over real-number arithmetic, proving all elementary functions are constructible from one primitive — a real-arithmetic analog to Boolean NAND completeness.

Projects

- **Smart Samarpan Academy** *Full-Stack EdTech Platform | 2026*
  React.js | Node.js | Express | MongoDB Atlas | Google Gemini API | Razorpay
  - Architected a full-stack AI-driven EdTech platform serving video courses, quizzes, and subscription management for Class 9–12 students.
  - Integrated **Google Gemini 1.5 Flash API** with custom prompt engineering to dynamically generate JEE/Olympiad-level quizzes and LaTeX formula sheets in under **1.5s**.
  - Implemented **Razorpay payment gateway** with server-side HMAC-SHA256 signature verification, preventing client-side payment tampering.
  - Built resilient backend middleware with exponential backoff retry logic for rate-limited API calls, achieving **99.9% uptime**.
- **Fake Review Detection System** *ML/NLP Web Application | 2026*
  Python | Flask | Scikit-learn | Gensim | NLTK | Google Gemini API
  - Built an end-to-end NLP pipeline classifying Amazon reviews as Real or Computer-Generated using Word2Vec embeddings and an **SVM classifier**.
  - Trained a **Word2Vec model (Gensim)** to generate 100-dimensional semantic embeddings combined with engineered linguistic features.
  - Developed a **Flask REST API** scraping live Amazon reviews via BeautifulSoup, with CAPTCHA-aware fallback for graceful degradation.
  - Integrated **Google Gemini API** to generate AI-powered summaries of review sets, improving user decision-making.

Technical Skills

Languages: C++, C, Python, JavaScript, HTML, CSS, SQL
Frameworks/Libraries: React.js, Node.js, Express, Flask
ML/NLP: Scikit-learn, Gensim, NLTK, Word2Vec
APIs & Integrations: Google Gemini API, Razorpay, BeautifulSoup
Databases/Cloud: MongoDB, MongoDB Atlas, MySQL
Developer Tools: Git, GitHub, VS Code, Postman
Core Competencies: Data Structures & Algorithms, OOP, REST API Development, Web Development, Responsive UI Design
Soft Skills: Leadership (led 3 projects), Teamwork (collaborated with 5+ team members), Problem-Solving (solved 50+ algorithmic problems), Communication

Achievements

- **NIMCET 2024** AIR 296 (Top 1%) among 25,000+ candidates | June 2024
- **Projects Delivered** Built 2+ web applications used by 100+ students during MCA coursework
- **GitHub Contributions** Contributed to 10+ repositories with 50+ stars combined
- **Frontend Development** Designed responsive UI components and interactive layouts for multiple projects
- **Competitive Programming** Solved **500+ problems** on platforms like **LeetCode** and **GeeksforGeeks**